Linux Commands & RAID

File system · permissions · processes · networking · RAID levels: exam-ready in 8 sheets

Linux
Commands (★ HIGH YIELD)

RAID
Levels 0/1/5/6/10 (★ HIGH YIELD)

Permissions
chmod / chown (★)

Core Commands
ls -la · pwd · cd · mkdir · rm -rf
grep -r · find · tail -f · head
chmod NNN · chown · ps aux
df -h · free -m · ip a · wget
sudo · kill · top · ssh · scp

RAID Table
RAID 0: stripe — N×disk, 0 fault
RAID 1: mirror — disk_size, 1 fault
RAID 5: (N-1)×disk, 1 fault ★
RAID 6: (N-2)×disk, 2 fault
RAID 10: N/2×disk, 1/pair ★

Legend: **yellow**=key term · ●=must-memorise · =memory trick · =exam trap

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Syllabus coverage: Chapter 18 OS syllabus covers Linux commands, file permissions, and RAID storage. This notebook covers all of it. Processes, scheduling, memory management, and deadlock have separate lessons.

Top 3 traps: ① chmod 701 = rwx-----x (0=---)
② RAID 5 usable = (N-1)×disk
③ RAID ≠ backup

(group=0!)
③ RAID 5 usable = (N-1)×disk
④ tail -f = real-time log
⑤ RAID ≠ backup
⑥ PHP is NOT a web server

1 File & Directory Commands ★ Most tested

Linux: is an open-source Unix-like OS kernel. It uses a **text-based CLI (terminal)** for administration. Commands follow: `command [options] [arguments]`. Case-sensitive.

ls -la
all files+perms

pwd
current dir

mkdir / rmdir
create / remove

rm -rf dir
delete recursive+for

cp / mv
copy / move

find . -name '*.txt'
recursive file search

Command	Flags	Purpose	Example
ls	-la	list all files (including hidden)	`ls -la /home`
pwd	—	print working directory	`pwd` → /home/user
cd	.. ~ /path	change directory; cd ~ → home	`cd /var/log`
mkdir	-p	create directory (and parents)	`mkdir -p a/b/c`
rm	-rf	remove recursively & force	`rm -rf /tmp/old`
cp / mv	-r	copy / move files	`cp -r src/ dst/`
cat	—	display file content	`cat /etc/hosts`
grep	-r -i -n	search text in files	`grep -rn 'error' /var/log/`
find	-name -type	recursive file search	`find . -name '*.c'`
wget / curl	-O	download from URL	`wget https://example.com/f.tar`
tar	-cvf / -xvf	archive / extract	`tar -cvf archive.tar dir/`

```
$ find . -name '*.[ch]' | xargs wc -l | tail -1
3456 total
→ counts all .c and .h lines recursively
```

```
$ ls -la /home/user
total 48 · -rw-r--r-- 1 user ... .bashrc
dotfile (.bashrc) = hidden, shown by -a
```

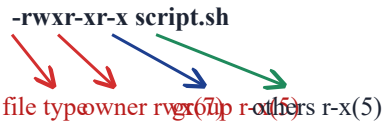
Memory trick: **PCLMFRG**: pwd · cd · ls · mkdir · find · rm · grep — the 7 daily commands every sysadmin types.

Worked example: Count lines in all .c and .h files: `find . -name '*.[ch]' | xargs wc -l | tail -1`

Exam trap: `ls -la` shows hidden files (dotfiles starting with **.**). `ls -l` does NOT.

2 File Permissions: chmod & chown ★ Highest-yield

Permission model: Every file has 3 user classes: **owner**, **group**, **others**. Each gets **r=4, w=2, x=1** bits. `chmod NNN file` — each digit = r×4+w×2+x×1.



Common chmod values

755 = rwxr-xr-x (executable/web dir)
644 = rw-r--r-- (regular files)
701 = rwx-----x (group=0, not --x!)
700 = rwx------ (private)
777 = rwxrwxrwx (dangerous!)
400 = r-------- (read-only)

Command	Purpose	Example
chmod 755 file	owner=rwx, group=r-x, others=r-x	standard for executables
chmod 644 file	owner=rw-, group=r--, others=r--	standard for data files
chmod 701 file	owner=rwx, group=---, others=--x	group gets NO bits (0=---)
chown user:group file	change owner and group	`chown www:www index.html`
chgrp group file	change group only	`chgrp dev project/`

Memory trick: r=**4**, w=**2**, x=**1** → add them: rwx=7, rw-=6, r-x=5, r--=4, --x=1, ---=0. Write the 3-digit octal left to right: owner · group · others.

Exam trap: chmod **701**: middle digit is **0** = **---** (no access for group), NOT **--x**. Many candidates write 701 as rwx--x--x — **WRONG**. 0 = ---, 1 = --x.

Worked example: `chmod 755 script.sh`: 7=rwx(owner), 5=r-x(group), 5=r-x(others) → **-rwxr-xr-x**. Standard for web server directories.

Exam trap: **chown** changes ownership. **chmod** changes permissions. They are separate operations — a common confusion in MCQs.

3 Process, Disk & Network Commands ★ PSU favourite

Process Management

Command	Purpose
ps aux	list all running processes
top	live CPU/memory usage
kill PID	terminate process by PID
kill -9 PID	SIGKILL — force terminate (uncatchable)
killall name	kill all processes by name
sudo command	run as root (audit trail)
su -	switch to root shell (not recommended)

Disk & Memory

Command	Purpose
df -h	disk space (human-readable)
du -sh /path	directory size
free -m	memory usage in MB
mount / umount	attach / detach filesystem

Viewing File Content

Command	Purpose
tail -n 10 file	last 10 lines
tail -f file	follow live log (real-time!)
head -n 10 file	first 10 lines
less file	paginated view (q to quit)
wc -l file	count lines in file

Network

Command	Purpose
ip a / ifconfig	show IP addresses
ping host	test connectivity
ssh user@host	remote login
scp file user@host:/path	secure file copy
wget URL	download file from URL
netstat / ss -tuln	show open ports

```
$ tail -f /var/log/syslog
Oct 30 10:05:12 server sshd[1234]: Accepted password for user
→ live stream — new lines appear as written. Ctrl+C to stop.
```

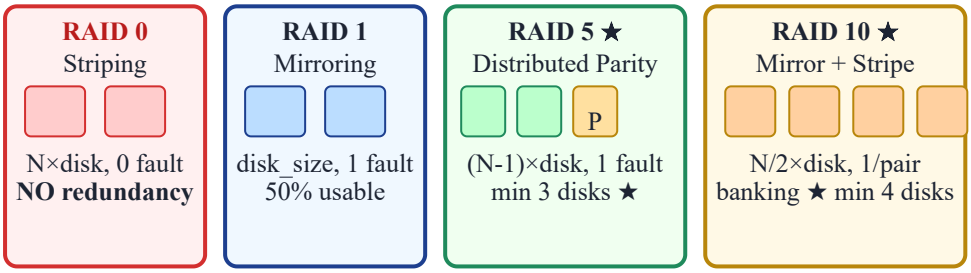
```
$ df -h
Filesystem Size Used Avail Use% Mounted on
/dev/sda1 50G 20G 28G 42% /
```

Memory trick: **tail -f** = "**f**ollow" = live log. **tail -n** = last N lines and exit. Exams ask which monitors a continuously updating log → **tail -f**.

Exam trap: Root login is disabled by default for security. Attackers must guess username AND password. `sudo` logs every action with originating username — **audit trail**.

4 RAID: Concepts & Levels ★ Bank/PSU favourite

RAID: = **Redundant Array of Independent Disks**. Combines multiple physical disks into one logical unit. Goals: **performance** (striping) or **reliability** (mirroring/parity) or both.



Level	Technique	Min disks	Fault tolerance	Usable capacity	Best for
RAID 0	Striping	2	**0**	N × disk	Speed (video editing)
RAID 1	Mirroring	2	1 drive	disk_size	OS drives, critical data
RAID 5	Dist. parity	3	1 drive	(N−1) × disk	File servers, NAS
RAID 6	Double parity	4	2 drives	(N−2) × disk	Large arrays
RAID 10	Mirror+stripe	4 (even)	1 per pair	N/2 × disk	Banking databases ★

RAID 5 — 4 drives × 2TB



→ (4-1)×2 = **6 TB** usable
1 drive = parity (distributed across all 4)
1 drive failure → rebuild from parity ✓

Memory trick: **RAID 5 usable = (N−1)×disk**. One disk "worth" of space goes to parity — distributed, but the math works as if one disk is consumed. Exams test this formula constantly.

Exam trap: **RAID ≠ backup.** RAID protects against **hardware disk failure** only — not against ransomware, accidental deletion, or data corruption (those get mirrored too!).

5 RAID Capacity Formulas ★ Numerical favourite

Formula sheet: Memorise these 5 formulas — they cover every RAID numerical in Bangladesh exams.

RAID	Usable capacity formula	4 × 2 TB example
RAID 0	`N × disk_size`	4 × 2 = **8 TB**
RAID 1	`disk_size` (one copy)	**2 TB** (regardless of N)
RAID 5	`(N − 1) × disk_size`	(4−1) × 2 = **6 TB**
RAID 6	`(N − 2) × disk_size`	(4−2) × 2 = **4 TB**
RAID 10	`(N / 2) × disk_size`	(4/2) × 2 = **4 TB**

Worked example 1: **BSCS AP 2026**: You have 4 drives of 2 TB each. Calculate usable storage for RAID 0, RAID 5, and RAID 10.

- RAID 0: 4 × 2 = **8 TB** (no redundancy — any failure loses all data)
- RAID 5: (4−1) × 2 = **6 TB** (1 drive for parity; tolerates 1 failure)
- RAID 10: (4/2) × 2 = **4 TB** (2 mirrored pairs striped together)

Worked example 2: **Combined Bank AME 2026**: 5 drives × 4 TB each in RAID 5. Why not 20 TB?

- Usable = (5−1) × 4 = **16 TB**
- 4 TB is consumed by distributed parity — one drive's *equivalent* across all 5
- Parity enables reconstruction if exactly ONE drive fails

Memory trick: For **banking** questions: which RAID? → **RAID 10** (mirror pairs that stripe = best I/O performance + redundancy, no parity calculation delay). RAID 5 is acceptable for file/archive servers.

Exam trap: RAID 10 needs an **even** number of disks (minimum 4). RAID 5 needs minimum 3. RAID 6 needs minimum 4. Confusing RAID 1 (min 2) with RAID 10 (min 4) is a common error.

Exam trap: RAID 5 tolerates exactly **1** disk failure. Two simultaneous failures → data lost. Use RAID 6 for 2-drive tolerance.

6 RAID Failure Recovery & Drive Types

RAID failure recovery procedure

1. **Controller detects failure** → array continues in **degraded mode** (using remaining disks + parity)
2. **Replace failed drive** with a new drive of equal or greater capacity

3. **Controller rebuilds** (reconstructs) lost data from parity/mirror onto new drive automatically
4. During rebuild: array is **vulnerable** — a second failure in RAID 5 will lose all data

Server types

Type	Description
Tower	Standalone, like desktop PC
Rack	19" rack, measured in U units
Blade	Thin modules in shared chassis, highest density

Data centre tiers

Tier	Uptime
Tier 1	99.671%
Tier 2	99.741%
Tier 3	99.982% — NCC Bangladesh
Tier 4	99.995% (fault tolerant)

Linux root login

Root login is **disabled by default** for security:
① Attacker must guess both **username AND password**
② `sudo` logs every privileged action with username
③ Prevents accidental destructive root commands
→ Use `sudo command` for single commands
→ Use `su -` for full root shell (not recommended)

PUE formula

PUE = Total facility power / IT equipment power
PUE = 1.0 = perfect (impossible)
Industry average ≈ 1.5
Green data centres: PUE < 1.2
Lower PUE = more efficient

Memory trick: Which RAID is best for a banking transaction server? → **RAID 10**: combines mirroring (full redundancy per pair) + striping (high throughput). No parity calculation penalty. Fast rebuild.

Exam trap: RAID is **NOT a backup**. A RAID array protects against hardware disk failure. Ransomware, accidental deletion, and fire destroy all copies simultaneously. Always have a separate backup.

7 Night-Before Sheet & Practice MCQs

Fact sheet:

- r=4, w=2, x=1; chmod NNN: owner·group·others
- chmod 755 → rwxr-xr-x; chmod 701 → rwx-----x
- ls -la shows hidden (dot) files
- tail -f = live/real-time log monitoring
- tail -n 10 = last 10 lines and exit
- find . -name '*.c' = recursive file search
- grep -r = recursive content search
- df -h = disk space; free -m = memory in MB
- RAID 0: N×disk, 0 fault (striping)
- RAID 1: disk_size, 1 fault (mirroring)
- RAID 5: (N-1)×disk, 1 fault (distributed parity, min 3)
- RAID 6: (N-2)×disk, 2 fault (double parity, min 4)
- RAID 10: N/2×disk, 1/pair (mirror+stripe, min 4)
- RAID ≠ backup; degraded mode on 1 failure
- PUE = total power / IT power (lower = better)
- Bangladesh NCC = Tier 3, Kaliakair, Gazipur

1. Which command monitors a continuously updating log file in real time?
a) tail -n 10 file b) head -f file **c) tail -f file** d) cat -r file

2. What is the permission string after `chmod 701 myfile`?
a) rwx-----x b) rwx--x--x c) rwxrwx--x d) rwx-wx--x

3. A RAID 5 array has 4 drives of 2 TB each. What is the usable storage?
a) 8 TB **b) 6 TB** c) 4 TB d) 2 TB

4. For a bank's core transaction server, which RAID is best?
a) RAID 0 b) RAID 5 c) RAID 1 **d) RAID 10**

5. Which Linux command shows disk space in human-readable format?

a) df -h b) du -h c) ls -h d) disk -h

6. RAID 0 uses 4 drives of 2 TB each. What is the total usable storage?

a) 8 TB b) 6 TB c) 4 TB d) 2 TB

7. Which RAID level provides fault tolerance through mirroring?

a) RAID 1 b) RAID 0 c) RAID 3 d) RAID 4

Answers: 1c 2a 3b 4d 5a 6a 7a

Written (5 marks): Explain RAID 0, 1, 5, and 10. Include usable capacity formula and fault tolerance for each. Which would you recommend for a Bangladesh bank's core transaction database and why?

Model answer:
RAID 0: N×disk, 0 fault — speed only, no redundancy.
RAID 1: disk_size, 1 fault — mirroring, 50% usable.
RAID 5: (N-1)×disk, 1 fault — distributed parity, min 3 disks, balance of space and safety.
RAID 10: N/2×disk, 1 per pair — mirror+stripe, min 4 disks.
Recommendation: **RAID 10** for banking — high I/O throughput (striping) + full redundancy (mirroring) + no parity calculation delay + fast rebuild. RAID 5 for less critical storage (reports, archives).